

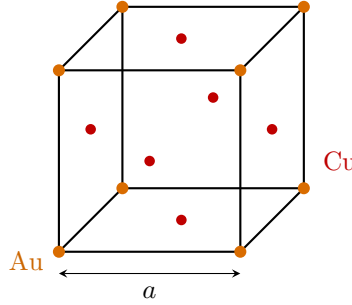
B6 Condensed-Matter Physics — Model Solutions, Trinity 2022

Question 1 — Cu_3Au structure factor and powder diffraction

Problem. Auricupride Cu_3Au , primitive cubic, $a = 0.375 \text{ nm}$, Au at corners, Cu at face centres. Derive structure factor; powder pattern at $\lambda = 0.154 \text{ nm}$; disordered alloy; partial-order parameter ρ .

(a) Structure factor and Bragg condition

(i) Unit cell.



Count Au per cell:

$$8 \times \frac{1}{8} = 1.$$

Count Cu per cell:

$$6 \times \frac{1}{2} = 3.$$

Stoichiometry Cu_3Au confirmed.

(ii) Structure factor. Basis: Au at $(0, 0, 0)$; Cu at $\frac{a}{2}(1, 1, 0)$, $\frac{a}{2}(1, 0, 1)$, $\frac{a}{2}(0, 1, 1)$. Reciprocal vector $\mathbf{G}_{hkl} = (2\pi/a)(h, k, l)$.

Definition:

$$S_{hkl} = \sum_{\alpha} f_{\alpha} e^{i\mathbf{G}_{hkl} \cdot \mathbf{r}_{\alpha}}.$$

Phase for Au at $(0, 0, 0)$:

$$\mathbf{G} \cdot \mathbf{r}_{\text{Au}} = 0.$$

Phase for Cu at $\frac{a}{2}(1, 1, 0)$:

$$\mathbf{G} \cdot \mathbf{r} = (2\pi/a)(h \cdot a/2 + k \cdot a/2) = \pi(h + k).$$

Phase for Cu at $\frac{a}{2}(1, 0, 1)$:

$$\mathbf{G} \cdot \mathbf{r} = \pi(h + l).$$

Phase for Cu at $\frac{a}{2}(0, 1, 1)$:

$$\mathbf{G} \cdot \mathbf{r} = \pi(k + l).$$

Insert basis:

$$S_{hkl} = f_{\text{Au}} + f_{\text{Cu}} [e^{i\pi(h+k)} + e^{i\pi(h+l)} + e^{i\pi(k+l)}]. \quad (1)$$

Identify α_{hkl} :

$$\alpha_{hkl} = e^{i\pi(h+k)} + e^{i\pi(h+l)} + e^{i\pi(k+l)}.$$

Each phase is ± 1 since $e^{i\pi n} = (-1)^n$. Unmixed (all even or all odd): every pair-sum is even:

$$\alpha_{hkl} = (+1) + (+1) + (+1) = +3.$$

Mixed parity: exactly two pair-sums are odd:

$$\alpha_{hkl} = (+1) + (-1) + (-1) = -1.$$

Combine:

$$\alpha_{hkl} = \begin{cases} +3, & h, k, l \text{ unmixed} \\ -1, & h, k, l \text{ mixed.} \end{cases} \quad (2)$$

(iii) Bragg's law. Elastic scattering:

$$|\mathbf{k}_f| = |\mathbf{k}_i| = 2\pi/\lambda.$$

Laue condition:

$$\mathbf{q} = \mathbf{k}_f - \mathbf{k}_i = \mathbf{G}_{hkl}.$$

Geometry of isosceles triangle:

$$|\mathbf{q}| = 2|\mathbf{k}_i| \sin \theta_{hkl}.$$

Substitute $|\mathbf{k}_i| = 2\pi/\lambda$:

$$G_{hkl} = \frac{4\pi \sin \theta_{hkl}}{\lambda}. \quad (3)$$

(b) Powder pattern at $\lambda = 0.154 \text{ nm}$

(i) Scattering angles. Cubic reciprocal magnitude:

$$G_{hkl} = \frac{2\pi}{a} \sqrt{h^2 + k^2 + l^2}.$$

Combine with (??):

$$\sin \theta_{hkl} = \frac{\lambda}{2a} \sqrt{h^2 + k^2 + l^2}.$$

Compute numerical prefactor:

$$\frac{\lambda}{2a} = \frac{0.154}{2 \times 0.375}.$$

Divide:

$$\frac{\lambda}{2a} = \frac{0.154}{0.750} = 0.2053.$$

For (110), $\sqrt{h^2 + k^2 + l^2} = \sqrt{2}$:

$$\sin \theta_{110} = 0.2053\sqrt{2} = 0.2904.$$

Invert sine:

$$\theta_{110} = \arcsin(0.2904) = 16.88^\circ.$$

Double the angle:

$$2\theta_{110} \approx 33.8^\circ.$$

For (111), $\sqrt{h^2 + k^2 + l^2} = \sqrt{3}$:

$$\sin \theta_{111} = 0.2053\sqrt{3} = 0.3556.$$

Invert sine:

$$\theta_{111} = \arcsin(0.3556) = 20.83^\circ.$$

Double the angle:

$$2\theta_{111} \approx 41.7^\circ.$$

(ii) Intensity ratio. Powder intensity (no LP, no geometry):

$$I_{hkl} = m_{hkl}|S_{hkl}|^2.$$

Multiplicities: $m_{110} = 12$, $m_{111} = 8$. From (??),(??): (110) is mixed, (111) unmixed:

$$S_{110} = f_{\text{Au}} - f_{\text{Cu}}, \quad S_{111} = f_{\text{Au}} + 3f_{\text{Cu}}. \quad (4)$$

Compute $(\sin \theta / \lambda)^2$ at (110) in nm^{-2} :

$$(\sin \theta_{110} / \lambda)^2 = (0.2904 / 0.154)^2.$$

Evaluate:

$$(\sin \theta_{110} / \lambda)^2 = (1.886)^2 = 3.555.$$

Same at (111):

$$(\sin \theta_{111} / \lambda)^2 = (0.3556 / 0.154)^2 = (2.309)^2 = 5.331.$$

Apply $f_X = Z_X e^{-b_X (\sin \theta / \lambda)^2}$. Cu at (110), exponent:

$$-b_{\text{Cu}} (\sin \theta / \lambda)^2 = -0.063 \times 3.555 = -0.2240.$$

Evaluate form factor:

$$f_{\text{Cu}}^{(110)} = 29 e^{-0.2240} = 23.18.$$

Au at (110), exponent:

$$-b_{\text{Au}} (\sin \theta / \lambda)^2 = -0.080 \times 3.555 = -0.2844.$$

Evaluate form factor:

$$f_{\text{Au}}^{(110)} = 79 e^{-0.2844} = 59.45.$$

Cu at (111), exponent:

$$-b_{\text{Cu}} (\sin \theta / \lambda)^2 = -0.063 \times 5.331 = -0.3358.$$

Evaluate form factor:

$$f_{\text{Cu}}^{(111)} = 29 e^{-0.3358} = 20.74.$$

Au at (111), exponent:

$$-b_{\text{Au}} (\sin \theta / \lambda)^2 = -0.080 \times 5.331 = -0.4265.$$

Evaluate form factor:

$$f_{\text{Au}}^{(111)} = 79 e^{-0.4265} = 51.59.$$

Substitute into S_{110} :

$$S_{110} = 59.45 - 23.18 = 36.27.$$

Substitute into S_{111} :

$$S_{111} = 51.59 + 3(20.74).$$

Multiply and add:

$$S_{111} = 51.59 + 62.22 = 113.81.$$

Form ratio:

$$\frac{I_{111}}{I_{110}} = \frac{m_{111}|S_{111}|^2}{m_{110}|S_{110}|^2} = \frac{8 \times (113.81)^2}{12 \times (36.27)^2}.$$

Evaluate squares:

$$\frac{I_{111}}{I_{110}} = \frac{8 \times 12953}{12 \times 1316}.$$

Multiply numerator and denominator:

$$\frac{I_{111}}{I_{110}} = \frac{103624}{15792}.$$

Divide:

$$\boxed{\frac{I_{111}}{I_{110}} \approx 6.6.}$$

(c) Fully disordered alloy

(i) **Intensities.** Average form factor at every site (1 Au + 3 Cu per cell, 4 sites):

$$\bar{f} = \frac{1}{4}f_{\text{Au}} + \frac{3}{4}f_{\text{Cu}}.$$

Replace $f_{\text{Au}}, f_{\text{Cu}} \rightarrow \bar{f}$ in (??):

$$S_{hkl} = \bar{f} [1 + e^{i\pi(h+k)} + e^{i\pi(h+l)} + e^{i\pi(k+l)}].$$

At (110), pair-sums are $h + k = 2$ (even), $h + l = 1$ (odd), $k + l = 1$ (odd):

$$S_{110} = \bar{f} [1 + e^{i2\pi} + e^{i\pi} + e^{i\pi}].$$

Evaluate exponentials:

$$S_{110} = \bar{f} (1 + 1 - 1 - 1) = 0.$$

Hence:

$$\boxed{S_{110} = 0, \quad I_{110} = 0.}$$

At (111), all three pair-sums are even (each = 2):

$$S_{111} = \bar{f} [1 + e^{i2\pi} + e^{i2\pi} + e^{i2\pi}].$$

Evaluate exponentials:

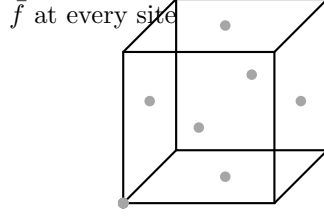
$$S_{111} = \bar{f} (1 + 1 + 1 + 1) = 4\bar{f}.$$

Substitute $\bar{f}$:

$$S_{111} = 4\left(\frac{1}{4}f_{\text{Au}} + \frac{3}{4}f_{\text{Cu}}\right) = f_{\text{Au}} + 3f_{\text{Cu}},$$

identical to part (b), so I_{111} unchanged.

(ii) **Extinction rules.** With one species $\bar{f}$ at corners and face centres, the structure is fcc with conventional parameter a . Fcc selection rule: h, k, l unmixed allowed, mixed forbidden. Hence (110) extinct; (111) allowed. Reflections such as (110) are *superlattice* lines: present only when ordering distinguishes corner and face sublattices.



(d) Partial ordering

(i) Site occupancies. Au probability at corner, linear in ρ between ordered ($\rho = 1$: $P = 1$) and random ($\rho = 0$: $P = \frac{1}{4}$):

$$P_{\text{Au}}^c = \rho \cdot 1 + (1 - \rho) \cdot \frac{1}{4}.$$

Simplify:

$$P_{\text{Au}}^c = \rho + \frac{1}{4} - \frac{1}{4}\rho = \frac{1}{4}(3\rho + 1).$$

Complement gives Cu at corner:

$$P_{\text{Cu}}^c = 1 - P_{\text{Au}}^c = 1 - \frac{1}{4}(3\rho + 1).$$

Simplify:

$$P_{\text{Cu}}^c = \frac{1}{4}(4 - 3\rho - 1) = \frac{3}{4}(1 - \rho).$$

Total-Au stoichiometry (1 corner site + 3 face sites supply 1 Au atom):

$$P_{\text{Au}}^c + 3P_{\text{Au}}^f = 1.$$

Solve for P_{Au}^f :

$$P_{\text{Au}}^f = \frac{1}{3}\left[1 - \frac{1}{4}(3\rho + 1)\right].$$

Combine over common denominator:

$$P_{\text{Au}}^f = \frac{1}{3} \cdot \frac{3-3\rho}{4} = \frac{1-\rho}{4}.$$

Complement gives Cu at face:

$$P_{\text{Cu}}^f = 1 - \frac{1-\rho}{4} = \frac{3+\rho}{4}.$$

Average corner form factor:

$$\langle f_{\text{corners}} \rangle = P_{\text{Au}}^c f_{\text{Au}} + P_{\text{Cu}}^c f_{\text{Cu}}.$$

Substitute:

$$\langle f_{\text{corners}} \rangle = \frac{1}{4}[(3\rho + 1)f_{\text{Au}} + 3(1 - \rho)f_{\text{Cu}}].$$

Average face form factor:

$$\langle f_{\text{faces}} \rangle = P_{\text{Au}}^f f_{\text{Au}} + P_{\text{Cu}}^f f_{\text{Cu}}.$$

Substitute:

$$\boxed{\langle f_{\text{faces}} \rangle = \frac{1}{4}[(1 - \rho)f_{\text{Au}} + (3 + \rho)f_{\text{Cu}}].} \quad (5)$$

(ii) Dependence on ρ . Replace $f_{\text{Au}} \rightarrow \langle f_c \rangle$ (corner) and $f_{\text{Cu}} \rightarrow \langle f_f \rangle$ (faces) in $S_{hkl} = f_{\text{Au}} + \alpha_{hkl}f_{\text{Cu}}$:

$$S_{hkl} = \langle f_c \rangle + \alpha_{hkl}\langle f_f \rangle.$$

For (111), $\alpha = +3$:

$$S_{111} = \frac{1}{4}[(3\rho + 1)f_{\text{Au}} + 3(1 - \rho)f_{\text{Cu}}] + \frac{3}{4}[(1 - \rho)f_{\text{Au}} + (3 + \rho)f_{\text{Cu}}].$$

Collect f_{Au} coefficient:

$$\text{coeff}_{f_{\text{Au}}} = \frac{1}{4}(3\rho + 1) + \frac{3}{4}(1 - \rho).$$

Combine over common denominator:

$$\text{coeff}_{f_{\text{Au}}} = \frac{1}{4}[(3\rho + 1) + 3(1 - \rho)].$$

Expand the bracket:

$$\text{coeff}_{f_{\text{Au}}} = \frac{1}{4}(3\rho + 1 + 3 - 3\rho).$$

Cancel ρ terms:

$$\text{coeff}_{f_{\text{Au}}} = \frac{1}{4} \cdot 4 = 1.$$

Collect f_{Cu} coefficient:

$$\text{coeff}_{f_{\text{Cu}}} = \frac{1}{4} \cdot 3(1 - \rho) + \frac{3}{4}(3 + \rho).$$

Combine over common denominator:

$$\text{coeff}_{f_{\text{Cu}}} = \frac{1}{4}[3(1 - \rho) + 3(3 + \rho)].$$

Expand the bracket:

$$\text{coeff}_{f_{\text{Cu}}} = \frac{1}{4}(3 - 3\rho + 9 + 3\rho).$$

Cancel ρ terms:

$$\text{coeff}_{f_{\text{Cu}}} = \frac{1}{4} \cdot 12 = 3.$$

Combine:

$$\boxed{S_{111} = f_{\text{Au}} + 3f_{\text{Cu}} \implies I_{111} \text{ independent of } \rho.} \quad (6)$$

For (110), $\alpha = -1$:

$$S_{110} = \frac{1}{4}[(3\rho + 1)f_{\text{Au}} + 3(1 - \rho)f_{\text{Cu}}] - \frac{1}{4}[(1 - \rho)f_{\text{Au}} + (3 + \rho)f_{\text{Cu}}].$$

Collect f_{Au} coefficient:

$$\text{coeff}_{f_{\text{Au}}} = \frac{1}{4}[(3\rho + 1) - (1 - \rho)].$$

Expand the bracket:

$$\text{coeff}_{f_{\text{Au}}} = \frac{1}{4}(3\rho + 1 - 1 + \rho).$$

Simplify:

$$\text{coeff}_{f_{\text{Au}}} = \frac{1}{4} \cdot 4\rho = \rho.$$

Collect f_{Cu} coefficient:

$$\text{coeff}_{f_{\text{Cu}}} = \frac{1}{4}[3(1 - \rho) - (3 + \rho)].$$

Expand the bracket:

$$\text{coeff}_{f_{\text{Cu}}} = \frac{1}{4}(3 - 3\rho - 3 - \rho).$$

Simplify:

$$\text{coeff}_{f_{\text{Cu}}} = \frac{1}{4}(-4\rho) = -\rho.$$

Combine:

$$S_{110} = \rho(f_{\text{Au}} - f_{\text{Cu}}).$$

Square modulus:

$$\boxed{I_{110} \propto |S_{110}|^2 = \rho^2(f_{\text{Au}} - f_{\text{Cu}})^2.} \quad (7)$$

Question 2 — NFE bands and the Peierls instability

Problem. Near zone boundary $E(y) = E_{\text{ZB}}(1 + y^2 \pm \sqrt{4y^2 + \epsilon^2})$, $y = 1 - ka/\pi$, $\epsilon = |V_G|/E_{\text{ZB}}$.

(a) Definitions and dispersion

(i) E_{ZB} . Free-electron energy at $k = \pi/a$:

$$\boxed{E_{\text{ZB}} = \frac{\hbar^2 \pi^2}{2m_e a^2}} \quad (8)$$

(ii) V_G and gap. Fourier expand the crystal potential:

$$U(x) = \sum_G V_G e^{iGx}.$$

V_G couples $|k\rangle$ and $|k - G\rangle$. At $k = \pi/a$, with $G = 2\pi/a$, the degenerate 2×2 block gives a gap:

$$\Delta E = 2|V_G|.$$

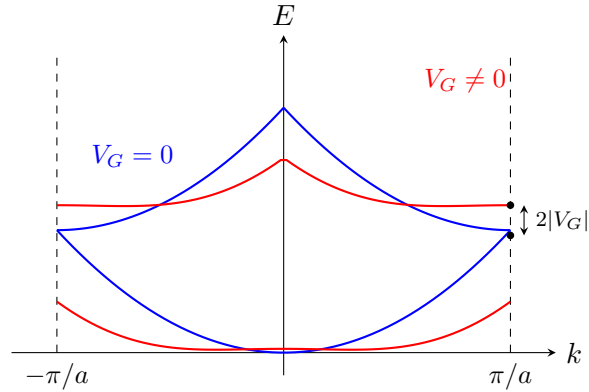
Check by setting $y = 0$ in given dispersion:

$$E_{\pm} = E_{\text{ZB}}(1 \pm \epsilon).$$

Subtract:

$$\Delta E = 2\epsilon E_{\text{ZB}} = 2|V_G|. \checkmark$$

(iii) Sketch.



(b) Energy of filled lower band

(i) **Total energy.** Lower band ($-$ sign), spin factor 2, density of states in k is $L/(2\pi)$:

$$\frac{E_{\text{tot}}}{L} = \frac{2}{2\pi} \int_{-\pi/a}^{\pi/a} E_{-}(k) dk.$$

Change variable $y = 1 - ka/\pi$:

$$dk = -(\pi/a)dy.$$

Limits: $k = -\pi/a \rightarrow y = 2$ and $k = \pi/a \rightarrow y = 0$; absorb the minus sign by swapping limits:

$$\frac{E_{\text{tot}}}{L} = \frac{1}{a} \int_0^2 E_{-}(y) dy.$$

The dispersion is symmetric about $y = 1$, so the integral over $[-1, 1]$ (as specified by the question) is the same length and gives:

$$\frac{E_{\text{tot}}}{L} = \frac{E_{\text{ZB}}}{a} \int_{-1}^1 (1 + y^2 - \sqrt{4y^2 + \epsilon^2}) dy. \quad (9)$$

Polynomial part:

$$\int_{-1}^1 (1 + y^2) dy = \left[y + \frac{1}{3} y^3 \right]_{-1}^1.$$

Evaluate:

$$\int_{-1}^1 (1 + y^2) dy = 2 + \frac{2}{3} = \frac{8}{3}.$$

Factor out 2 from the square root:

$$\sqrt{4y^2 + \epsilon^2} = 2\sqrt{y^2 + (\epsilon/2)^2}.$$

Insert into the integral:

$$\int_{-1}^1 \sqrt{4y^2 + \epsilon^2} dy = 2 \int_{-1}^1 \sqrt{y^2 + (\epsilon/2)^2} dy.$$

Apply the given primitive with $c = \epsilon/2$:

$$= 2 \left[\frac{1}{2} y \sqrt{(\epsilon/2)^2 + y^2} + \frac{1}{2} (\epsilon/2)^2 \log(y + \sqrt{(\epsilon/2)^2 + y^2}) \right]_{-1}^1.$$

Absorb the outer 2 into the bracket:

$$= \left[y \sqrt{(\epsilon/2)^2 + y^2} + (\epsilon/2)^2 \log(y + \sqrt{(\epsilon/2)^2 + y^2}) \right]_{-1}^1.$$

Evaluate the surd at $y = \pm 1$:

$$\sqrt{(\epsilon/2)^2 + 1} = \sqrt{\frac{\epsilon^2 + 4}{4}} = \frac{1}{2} \sqrt{4 + \epsilon^2}.$$

Surd contribution (top minus bottom):

$$\left[y \sqrt{(\epsilon/2)^2 + y^2} \right]_{-1}^1 = \frac{1}{2} \sqrt{4 + \epsilon^2} - (-1) \cdot \frac{1}{2} \sqrt{4 + \epsilon^2}.$$

Simplify:

$$\left[y \sqrt{(\epsilon/2)^2 + y^2} \right]_{-1}^1 = \sqrt{4 + \epsilon^2}.$$

Log contribution, using $c^2 = \epsilon^2/4$:

$$\left[(\epsilon/2)^2 \log(y + \sqrt{(\epsilon/2)^2 + y^2}) \right]_{-1}^1 = \frac{\epsilon^2}{4} \log \frac{1 + \frac{1}{2} \sqrt{4 + \epsilon^2}}{-1 + \frac{1}{2} \sqrt{4 + \epsilon^2}}.$$

Multiply numerator and denominator by 2:

$$= \frac{\epsilon^2}{4} \log \frac{2 + \sqrt{4 + \epsilon^2}}{\sqrt{4 + \epsilon^2} - 2}.$$

Assemble:

$$\boxed{\frac{E_{\text{tot}}}{L} = \frac{E_{\text{ZB}}}{a} \left[\frac{8}{3} - \sqrt{4 + \epsilon^2} - \frac{\epsilon^2}{4} \log \frac{2 + \sqrt{4 + \epsilon^2}}{\sqrt{4 + \epsilon^2} - 2} \right]}. \quad (10)$$

(ii) Monotonic decrease. Define the bracket subtracted from $8/3$:

$$u(\epsilon) = \sqrt{4 + \epsilon^2} + \frac{\epsilon^2}{4} \log \frac{2 + \sqrt{4 + \epsilon^2}}{\sqrt{4 + \epsilon^2} - 2}.$$

Let $s = \sqrt{4 + \epsilon^2}$; differentiate the surd:

$$\frac{ds}{d\epsilon} = \frac{\epsilon}{s}.$$

Differentiate $\mathcal{L} = \log \frac{s+2}{s-2}$ using the chain rule:

$$\frac{d\mathcal{L}}{d\epsilon} = \frac{ds}{d\epsilon} \left(\frac{1}{s+2} - \frac{1}{s-2} \right).$$

Combine the fractions:

$$\frac{1}{s+2} - \frac{1}{s-2} = \frac{(s-2) - (s+2)}{s^2 - 4} = \frac{-4}{s^2 - 4}.$$

Use $s^2 - 4 = \epsilon^2$:

$$\frac{d\mathcal{L}}{d\epsilon} = \frac{\epsilon}{s} \cdot \frac{-4}{\epsilon^2} = \frac{-4}{s\epsilon}.$$

Differentiate $u = s + (\epsilon^2/4)\mathcal{L}$ by product rule:

$$\frac{du}{d\epsilon} = \frac{ds}{d\epsilon} + \frac{2\epsilon}{4}\mathcal{L} + \frac{\epsilon^2}{4} \cdot \frac{d\mathcal{L}}{d\epsilon}.$$

Substitute the derivatives:

$$\frac{du}{d\epsilon} = \frac{\epsilon}{s} + \frac{\epsilon}{2}\mathcal{L} + \frac{\epsilon^2}{4} \cdot \frac{-4}{s\epsilon}.$$

Simplify the last term:

$$\frac{du}{d\epsilon} = \frac{\epsilon}{s} + \frac{\epsilon}{2}\mathcal{L} - \frac{\epsilon}{s}.$$

Cancel the ϵ/s terms:

$$\frac{du}{d\epsilon} = \frac{\epsilon}{2} \log \frac{2 + \sqrt{4 + \epsilon^2}}{\sqrt{4 + \epsilon^2} - 2}.$$

Argument of log > 1 for $\epsilon > 0$ (numerator $>$ denominator), so $du/d\epsilon \geq 0$:

$$u(\epsilon) \geq u(0) = \sqrt{4} = 2.$$

Hence the bracket $8/3 - u(\epsilon)$ is largest at $\epsilon = 0$:

$$E_{\text{tot}}(\epsilon) \leq E_{\text{tot}}(0).$$

Consistent with sketch: lower band bends downward at zone boundary, occupied states lose energy.

(iii) Small- ϵ expansion. Factor out 2 from the surd:

$$\sqrt{4 + \epsilon^2} = 2\sqrt{1 + \epsilon^2/4}.$$

Binomial expand for $\epsilon \ll 1$:

$$\sqrt{4 + \epsilon^2} \simeq 2\left(1 + \frac{1}{2} \cdot \frac{\epsilon^2}{4}\right) = 2 + \frac{\epsilon^2}{4}.$$

Numerator of log argument:

$$2 + \sqrt{4 + \epsilon^2} \simeq 4 + \frac{\epsilon^2}{4} \simeq 4.$$

Denominator of log argument:

$$\sqrt{4 + \epsilon^2} - 2 \simeq \frac{\epsilon^2}{4}.$$

Form the ratio:

$$\frac{2 + \sqrt{4 + \epsilon^2}}{\sqrt{4 + \epsilon^2} - 2} \simeq \frac{4}{\epsilon^2/4} = \frac{16}{\epsilon^2}.$$

Apply $\log(16/\epsilon^2) = 2 \log(4/\epsilon)$:

$$\frac{\epsilon^2}{4} \log \frac{16}{\epsilon^2} = \frac{\epsilon^2}{4} \cdot 2 \log \frac{4}{\epsilon} = \frac{\epsilon^2}{2} \log \frac{4}{\epsilon}.$$

Rewrite using $\log(4/\epsilon) = -\log(\epsilon/4)$:

$$\frac{\epsilon^2}{4} \log \frac{16}{\epsilon^2} = -\frac{\epsilon^2}{2} \log \frac{\epsilon}{4}.$$

Insert into (??) (note minus sign on the log term):

$$\frac{E_{\text{tot}}}{L} \simeq \frac{E_{\text{ZB}}}{a} \left[\frac{8}{3} - \left(2 + \frac{\epsilon^2}{4} \right) - \left(-\frac{\epsilon^2}{2} \log \frac{\epsilon}{4} \right) \right].$$

Simplify the signs:

$$\frac{E_{\text{tot}}}{L} \simeq \frac{E_{\text{ZB}}}{a} \left[\frac{8}{3} - 2 - \frac{\epsilon^2}{4} + \frac{\epsilon^2}{2} \log \frac{\epsilon}{4} \right].$$

Combine the constants:

$$\frac{8}{3} - 2 = \frac{2}{3}.$$

Substitute:

$$\frac{E_{\text{tot}}}{L} \simeq \frac{E_{\text{ZB}}}{a} \left[\frac{2}{3} - \frac{\epsilon^2}{4} + \frac{\epsilon^2}{2} \log \frac{\epsilon}{4} \right].$$

Factor out $\epsilon^2/4$ from the last two terms:

$$\frac{E_{\text{tot}}}{L} \simeq \frac{E_{\text{ZB}}}{a} \left[\frac{2}{3} - \frac{\epsilon^2}{4} \left(1 - 2 \log \frac{\epsilon}{4} \right) \right].$$

Read off:

$$\boxed{\alpha = 1, \quad \beta = 2.} \tag{11}$$

(c) Dimerised chain

(i) V_G **and gap**. Expand the given potential:

$$V(x) = V_0 \cos(4\pi x/a) - 2\pi\delta V_0 \cos(2\pi x/a) + 2\pi\delta V_0 \cos(6\pi x/a).$$

Use Euler:

$$\cos(2\pi n x/a) = \frac{1}{2} (e^{i2\pi n x/a} + e^{-i2\pi n x/a}).$$

Read off V_{G_1} ($G_1 = 2\pi/a$, from $-2\pi\delta V_0 \cos(2\pi x/a)$):

$$V_{G_1} = -2\pi\delta V_0 \cdot \frac{1}{2} = -\pi\delta V_0.$$

Read off V_{G_2} ($G_2 = 4\pi/a$, from $V_0 \cos(4\pi x/a)$):

$$V_{G_2} = V_0 \cdot \frac{1}{2} = \frac{1}{2} V_0.$$

Read off V_{G_3} ($G_3 = 6\pi/a$, from $+2\pi\delta V_0 \cos(6\pi x/a)$):

$$V_{G_3} = +2\pi\delta V_0 \cdot \frac{1}{2} = +\pi\delta V_0.$$

Two atoms per cell of size a , BZ is $|k| \leq \pi/a$, half-filling fills lower band up to $k = \pi/a$. Gap at zone boundary uses $G_1 = 2\pi/a$:

$$\Delta E = 2|V_{G_1}|.$$

Substitute:

$$\Delta E = 2\pi|\delta|V_0.$$

Boxed result:

$$\boxed{\Delta E = 2\pi|\delta|V_0 \propto |\delta|}. \quad (12)$$

At $\delta = 0$: $V_{G_1} = 0 \Rightarrow \Delta E = 0$.

(ii) No gap at $\delta = 0$. Potential reduces to $V_0 \cos(4\pi x/a)$, period $a/2$. True lattice constant $a/2$, BZ $|k| \leq 2\pi/a$, half-filled $\Rightarrow$ metallic. Gap from V_{G_2} opens at $k = 2\pi/a$, above E_F . Dimerising doubles the period: zone folds, the band-touching at π/a acquires gap $\propto \delta$.

(iii) Energy density. Define $\epsilon(\delta)$ from $|V_{G_1}| = \pi|\delta|V_0$:

$$\epsilon(\delta) = \frac{|V_{G_1}|}{E_{\text{ZB}}} = \frac{\pi V_0}{E_{\text{ZB}}} |\delta| = A|\delta|, \quad A = \frac{\pi V_0}{E_{\text{ZB}}}.$$

Insert $\epsilon = A|\delta|$ into the small- ϵ result of part (b)(iii):

$$\boxed{\frac{E_{\text{tot}}(\delta)}{L} = \frac{E_{\text{ZB}}}{a} \left[\frac{2}{3} - \frac{A^2 \delta^2}{4} \left(1 - 2 \log \frac{A|\delta|}{4} \right) \right]}. \quad (13)$$

Isolate the δ -dependent electronic part:

$$E_{\text{el}}(\delta) = \frac{E_{\text{ZB}}}{a} \left[-\frac{A^2 \delta^2}{4} + \frac{A^2 \delta^2}{2} \log \frac{A|\delta|}{4} \right].$$

As $\delta \rightarrow 0$, $\log(A|\delta|/4) \rightarrow -\infty$, so the second term dominates. Rewrite using $\log(A|\delta|/4) = \log A - \log 4 + \log |\delta| = -\log(1/|\delta|) + \text{const}$:

$$E_{\text{el}}(\delta) \simeq -C\delta^2 \log \frac{1}{|\delta|}, \quad C = \frac{A^2 E_{\text{ZB}}}{2a} > 0.$$

(d) Peierls instability

Sum elastic and electronic contributions:

$$E_{\text{tot}}(\delta) = \kappa\delta^2 - C\delta^2 \log \frac{1}{|\delta|} + \text{const}.$$

Compare leading scales as $\delta \rightarrow 0$:

$$\kappa\delta^2 \sim \delta^2, \quad C\delta^2 \log(1/|\delta|) \sim \delta^2 \log(1/|\delta|).$$

The logarithm diverges as $\delta \rightarrow 0$, so for any finite $\kappa > 0$ there exists $|\delta| > 0$ small enough that:

$$C \log(1/|\delta|) > \kappa.$$

Multiply by δ^2 :

$$C\delta^2 \log(1/|\delta|) > \kappa\delta^2.$$

Therefore at small but finite $|\delta|$:

$$E_{\text{tot}}(\delta) < E_{\text{tot}}(0).$$

The minimum sits at $\delta^* \neq 0$:

$$\boxed{\text{minimum at } \delta^* \neq 0 \Rightarrow \text{spontaneous dimerisation (Peierls)}}.$$

Question 3 — Semiconductor doping and activation curves

Problem. Derive law of mass action; donor occupation; low- T electron concentration; read N_D , E_i , E_g from $n(T)$ plots.

(a) Law of mass action

Non-degenerate semiconductor: μ lies in the gap, so $\beta(\epsilon - \mu) \gg 1$ throughout the conduction band. Fermi-Dirac reduces to Boltzmann:

$$f_c(\epsilon) = \frac{1}{e^{\beta(\epsilon - \mu)} + 1} \simeq e^{-\beta(\epsilon - \mu)}.$$

Substitute into the given $n(T)$:

$$n(T) = \frac{(2m_e)^{3/2}}{2\pi^2\hbar^3} \int_{\epsilon_c}^{\infty} (\epsilon - \epsilon_c)^{1/2} e^{-\beta(\epsilon - \mu)} d\epsilon.$$

Split the exponent: $\epsilon - \mu = (\epsilon - \epsilon_c) + (\epsilon_c - \mu)$:

$$n(T) = \frac{(2m_e)^{3/2}}{2\pi^2\hbar^3} e^{-\beta(\epsilon_c - \mu)} \int_{\epsilon_c}^{\infty} (\epsilon - \epsilon_c)^{1/2} e^{-\beta(\epsilon - \epsilon_c)} d\epsilon.$$

Change variable $\xi = \beta(\epsilon - \epsilon_c)$, $d\epsilon = d\xi/\beta$:

$$n(T) = \frac{(2m_e)^{3/2}}{2\pi^2\hbar^3} \beta^{-3/2} e^{-\beta(\epsilon_c - \mu)} \int_0^{\infty} \xi^{1/2} e^{-\xi} d\xi.$$

Apply given integral $\int_0^{\infty} \xi^{1/2} e^{-\xi} d\xi = \sqrt{\pi}/2$:

$$n(T) = \frac{(2m_e)^{3/2}}{2\pi^2\hbar^3} (k_B T)^{3/2} \frac{\sqrt{\pi}}{2} e^{-\beta(\epsilon_c - \mu)}.$$

Tidy the prefactor into $N_c(T) \equiv 2(m_e k_B T / 2\pi\hbar^2)^{3/2}$:

$$n(T) = 2 \left(\frac{m_e k_B T}{2\pi\hbar^2} \right)^{3/2} e^{-\beta(\epsilon_c - \mu)} \equiv N_c(T) e^{-\beta(\epsilon_c - \mu)}. \quad (14)$$

Repeat for holes using $f_v = 1 - f_{\text{FD}} \simeq e^{-\beta(\mu - \epsilon)}$ in the valence band:

$$p(T) = 2 \left(\frac{m_h k_B T}{2\pi\hbar^2} \right)^{3/2} e^{-\beta(\mu - \epsilon_v)} \equiv N_v(T) e^{-\beta(\mu - \epsilon_v)}. \quad (15)$$

Multiply (??),(??); the μ exponents cancel and $(\epsilon_c - \epsilon_v) = E_g$:

$$n(T)p(T) = N_c(T)N_v(T) e^{-\beta(\epsilon_c - \epsilon_v)}.$$

Substitute explicit prefactors:

$$\boxed{n(T)p(T) = 4 \left(\frac{k_B T}{2\pi\hbar^2} \right)^3 (m_e m_h)^{3/2} e^{-\beta E_g}.} \quad (16)$$

Approximation used: Boltzmann tails (non-degenerate carriers).

(b) Donor-doped semiconductor

(i) Donor occupation. Single localised level: empty ($N = 0$, $E = 0$) or singly occupied ($N = 1$, $E = \epsilon_D$). Grand partition function:

$$Z_1 = 1 + e^{-\beta(\epsilon_D - \mu)}.$$

Probability of occupied state ($N = 1$):

$$P_{\text{full}} = \frac{e^{-\beta(\epsilon_D - \mu)}}{Z_1}.$$

Probability of empty state ($N = 0$):

$$P_{\text{empty}} = \frac{1}{Z_1}.$$

Form ratio:

$$\frac{n_{\text{full}}}{n_{\text{empty}}} = \frac{P_{\text{full}}}{P_{\text{empty}}} = e^{-\beta(\epsilon_D - \mu)}.$$

Flip the sign in the exponent:

$$\boxed{\frac{n_{\text{full}}}{n_{\text{empty}}} = e^{\beta(\mu - \epsilon_D)}}. \quad (17)$$

(ii) Low- T electron concentration. Charge neutrality with $k_B T \ll E_g$ (valence-band excitation negligible): conduction electrons come from ionised donors. Each ionised donor is empty, each unionised donor is full:

$$n = n_{\text{empty}}, \quad N_D - n = n_{\text{full}}.$$

Insert into (??):

$$\frac{N_D - n}{n} = e^{\beta(\mu - \epsilon_D)}.$$

Solve for $e^{\beta\mu}$:

$$e^{\beta\mu} = \frac{N_D - n}{n} e^{\beta\epsilon_D}.$$

From (??), $n = N_c e^{-\beta(\epsilon_c - \mu)}$:

$$e^{\beta\mu} = \frac{n}{N_c} e^{\beta\epsilon_c}.$$

Equate the two expressions for $e^{\beta\mu}$:

$$\frac{n}{N_c} e^{\beta\epsilon_c} = \frac{N_D - n}{n} e^{\beta\epsilon_D}.$$

Cross-multiply:

$$\frac{n^2}{N_D - n} = N_c(T) e^{-\beta(\epsilon_c - \epsilon_D)}.$$

Substitute $E_i = \epsilon_c - \epsilon_D$:

$$\frac{n^2}{N_D - n} = N_c(T) e^{-\beta E_i}.$$

Use the explicit $N_c = 2(m_e/2\pi\beta\hbar^2)^{3/2}$:

$$\boxed{\frac{n^2}{N_D - n} = 2 \left(\frac{m_e}{2\pi\beta\hbar^2} \right)^{3/2} e^{-\beta E_i}}. \quad (18)$$

(c) Reading the activation curves

(i) Three regimes.

- *Freeze-out* (low T , large $1/T$): donors only partially ionising. Take dilute limit $n \ll N_D$ in (??): $n^2/N_D \simeq N_c e^{-\beta E_i}$, hence $n \simeq \sqrt{N_D N_c} e^{-E_i/(2k_B T)}$. Slope of $\ln n$ vs $1/T$ is $-E_i/(2k_B)$.
- *Saturation* (intermediate T): $k_B T \gg E_i$ so all donors ionised, while $k_B T \ll E_g$ so no intrinsic carriers. Plateau at $n \simeq N_D$.
- *Intrinsic* (high T , small $1/T$): thermal pair generation across E_g exceeds N_D . Charge neutrality $n = p$ gives $n \simeq n_i = \sqrt{np} \propto T^{3/2} e^{-E_g/(2k_B T)}$, slope $-E_g/(2k_B)$.

(ii) Numerical values.

Saturation plateau on $\log_{10} n$ vs $1/T$ at $\log_{10} n \approx 18$:

$$N_D \approx 1 \times 10^{18} \text{ m}^{-3}.$$

Read freeze-out slope from plot ($1/T \in [0.10, 0.30] \text{ K}^{-1}$, $\log_{10} n$ runs ~ 17 down to ~ 13.5):

$$\frac{\Delta \log_{10} n}{\Delta(1/T)} \approx \frac{13.5 - 17}{0.30 - 0.10}.$$

Evaluate the quotient:

$$\frac{\Delta \log_{10} n}{\Delta(1/T)} \approx \frac{-3.5}{0.20} = -17.5 \text{ K}.$$

Convert to natural-log slope using $\ln n = \ln 10 \cdot \log_{10} n$:

$$\text{slope}_{\ln n} = \ln 10 \times (-17.5).$$

Evaluate:

$$\text{slope}_{\ln n} = 2.303 \times (-17.5) \approx -40.3 \text{ K}.$$

Identify $\text{slope}_{\ln n} = -E_i/(2k_B)$:

$$E_i = 2k_B \times 40 \text{ K}.$$

Substitute $k_B = 8.617 \times 10^{-5} \text{ eV/K}$:

$$E_i = 2 \times (8.617 \times 10^{-5}) \times 40 \text{ eV}.$$

Evaluate:

$$E_i \approx 7 \text{ meV}.$$

Read intrinsic slope (high- T tail, $\log_{10} n \approx 18$ at $1/T \approx 0.02 \text{ K}^{-1}$ rising to $\log_{10} n \approx 23$ at $1/T \approx 0.002 \text{ K}^{-1}$):

$$\frac{\Delta \log_{10} n}{\Delta(1/T)} \approx \frac{23 - 18}{0.002 - 0.020}.$$

Evaluate the quotient:

$$\frac{\Delta \log_{10} n}{\Delta(1/T)} \approx \frac{5}{-0.018} \approx -278 \text{ K}.$$

Convert to natural-log slope:

$$\text{slope}_{\ln n} = \ln 10 \times (-278) \approx -640 \text{ K}.$$

Identify with $-E_g/(2k_B)$:

$$E_g = 2k_B \times 640 \text{ K}.$$

Substitute k_B :

$$E_g = 2 \times (8.617 \times 10^{-5}) \times 640 \text{ eV}.$$

Evaluate:

$$E_g \approx 0.11 \text{ eV}.$$

Question 4 — Hund's rules and Curie–Weiss for $\text{PrO}_{1.5}$

Problem. State Hund's rules; derive Curie–Weiss; extract m_{eff} , Θ from $\text{PrO}_{1.5}$ data; compute Landé $\tilde{g}$ for Pr^{3+} .

(a) Hund's rules

1. Maximise S subject to Pauli.
2. Maximise L subject to chosen S .
3. $J = |L - S|$ if shell less than half-filled, $J = L + S$ if more.

Origin. Rules 1, 2: electron–electron Coulomb repulsion. Parallel spins force antisymmetric spatial wavefunction $\Rightarrow$ node when electrons coincide $\Rightarrow$ smaller repulsion. Larger L keeps electrons orbiting in same sense, also reducing overlap. Rule 3: spin–orbit coupling $\lambda \mathbf{L} \cdot \mathbf{S}$; $\lambda > 0$ for $<$ half-filled (favours $J = |L - S|$), $\lambda < 0$ for $>$ half-filled (favours $J = L + S$, hole picture).

(b) Curie–Weiss

(i) Derivation. Quote Curie law for volume susceptibility (Brillouin function expanded for $\mu_B B \ll k_B T$):

$$\chi_{\text{Curie}} = \frac{\mu_0 n \tilde{g}^2 \mu_B^2 J(J+1)}{3k_B T}.$$

Convert to molar by multiplying by molar volume V_m and using $nV_m = N_A$:

$$\chi_{\text{mol}} = \chi_{\text{Curie}} V_m = \frac{\mu_0 N_A \tilde{g}^2 \mu_B^2 J(J+1)}{3k_B T}.$$

Define $m_{\text{eff}} = \tilde{g} \sqrt{J(J+1)}$:

$$\chi_{\text{mol}} = \frac{\mu_0 N_A \mu_B^2}{3k_B T} m_{\text{eff}}^2.$$

Evaluate prefactor $\mu_0 N_A \mu_B^2 / (3k_B)$ in SI:

$$\frac{\mu_0 N_A \mu_B^2}{3k_B} = \frac{(4\pi \times 10^{-7})(6.022 \times 10^{23})(9.274 \times 10^{-24})^2}{3(1.381 \times 10^{-23})}.$$

Compute μ_B^2 :

$$\mu_B^2 = (9.274 \times 10^{-24})^2 = 8.601 \times 10^{-47}.$$

Substitute the squared value:

$$\frac{\mu_0 N_A \mu_B^2}{3k_B} = \frac{(1.257 \times 10^{-6})(6.022 \times 10^{23})(8.601 \times 10^{-47})}{4.143 \times 10^{-23}}.$$

Multiply the numerator:

$$\text{num} = 1.257 \times 6.022 \times 8.601 \times 10^{-30} = 6.51 \times 10^{-29}.$$

Divide by denominator:

$$\frac{\mu_0 N_A \mu_B^2}{3k_B} \approx 1.57 \times 10^{-6} \text{ m}^3 \text{ K mol}^{-1}.$$

Apply mean-field (Weiss) shift $T \rightarrow T - \Theta$:

$$\chi_{\text{mol}}(T) = (1.57 \times 10^{-6} \text{ m}^3 \text{ K mol}^{-1}) \frac{m_{\text{eff}}^2}{T - \Theta}. \quad (19)$$

(ii) Landé relation.

$$m_{\text{eff}} = \tilde{g} \sqrt{J(J+1)}.$$

(iii) Θ . Intercept of $1/\chi$ vs T on the T -axis. $\Theta > 0$: ferromagnetic interactions; $\Theta < 0$: antiferromagnetic.

(c) Extracting m_{eff} and Θ

Read two points from the line: $(T, 1/\chi) \approx (50, 7)$ and $(300, 17)$ in units of 10^6 mol/m^3 .

Slope of $1/\chi_{\text{mol}}$ vs T :

$$\text{slope} = \frac{(17 - 7) \times 10^6}{(300 - 50) \text{ K}}.$$

Evaluate:

$$\text{slope} = \frac{10 \times 10^6}{250} = 4.0 \times 10^4 \text{ mol m}^{-3} \text{ K}^{-1}.$$

From (??), invert: $1/\chi_{\text{mol}} = (T - \Theta)/[1.57 \times 10^{-6} m_{\text{eff}}^2]$, so the slope satisfies:

$$\text{slope} = \frac{1}{1.57 \times 10^{-6} m_{\text{eff}}^2}.$$

Solve for m_{eff}^2 :

$$m_{\text{eff}}^2 = \frac{1}{(1.57 \times 10^{-6})(4.0 \times 10^4)}.$$

Evaluate denominator:

$$(1.57 \times 10^{-6})(4.0 \times 10^4) = 6.28 \times 10^{-2}.$$

Take reciprocal:

$$m_{\text{eff}}^2 = \frac{1}{6.28 \times 10^{-2}} = 15.92.$$

Square root:

$$m_{\text{eff}} \approx 3.99 \mu_B.$$

Find Θ from the linear form $1/\chi = \text{slope} \cdot (T - \Theta)$ at the point $(50, 7 \times 10^6)$:

$$\Theta = T - \frac{1/\chi}{\text{slope}} \Big|_{(50, 7 \times 10^6)}.$$

Substitute the numbers:

$$\Theta = 50 \text{ K} - \frac{7 \times 10^6}{4.0 \times 10^4} \text{ K}.$$

Evaluate the fraction:

$$\frac{7 \times 10^6}{4.0 \times 10^4} = 175 \text{ K}.$$

Subtract:

$$\Theta = (50 - 175) \text{ K} = -125 \text{ K}.$$

Negative Θ indicates AFM coupling:

$$\Theta \approx -125 \text{ K} \Rightarrow \text{antiferromagnetic.}$$

(d) Pr^{3+} ground term

(i) L, S, J . Configuration $4f^2$, $\ell = 3$ (seven $m_\ell \in \{-3, \dots, 3\}$). Hund 1: parallel spins:

$$S = \frac{1}{2} + \frac{1}{2} = 1.$$

Hund 2: occupy $m_\ell = +3, +2$:

$$L = 3 + 2 = 5 \quad (\text{H term}).$$

Hund 3: less than half-filled (2 of 14):

$$J = |L - S| = 5 - 1 = 4.$$

Ground term:

$$^3H_4.$$

(ii) Landé factor. Use formula with $S = 1$, $L = 5$, $J = 4$:

$$\tilde{g}_{\text{th}} = \frac{3}{2} + \frac{S(S+1) - L(L+1)}{2J(J+1)}.$$

Compute the products: $S(S+1) = 2$, $L(L+1) = 30$, $J(J+1) = 20$:

$$\tilde{g}_{\text{th}} = \frac{3}{2} + \frac{2 - 30}{2 \times 20}.$$

Evaluate the fraction:

$$\tilde{g}_{\text{th}} = \frac{3}{2} + \frac{-28}{40} = \frac{3}{2} - \frac{7}{10}.$$

Combine:

$$\tilde{g}_{\text{th}} = 1.5 - 0.7 = 0.80.$$

Theoretical effective moment:

$$m_{\text{eff,th}} = \tilde{g}_{\text{th}} \sqrt{J(J+1)} = 0.80 \sqrt{20}.$$

Evaluate:

$$m_{\text{eff,th}} = 0.80 \times 4.472 = 3.58 \mu_B.$$

Experimental Landé from (c) using $m_{\text{eff,exp}} = 3.99 \mu_B$:

$$\tilde{g}_{\text{exp}} = \frac{m_{\text{eff,exp}}}{\sqrt{J(J+1)}} = \frac{3.99}{4.472}.$$

Divide:

$$\tilde{g}_{\text{exp}} \approx 0.89.$$

$\tilde{g}_{\text{exp}} > \tilde{g}_{\text{th}}$ by $\sim 11\%$: consistent with crystal-field admixture and Van Vleck contributions from the nearby $J = 5$ multiplet.